Arpit Agarwal

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SUMMARY

PhD roboticist with 10+ years across reinforcement learning, GPU simulation, and production manipulation. At Dexterity I lead GPU simulation and RL, training packing policies that run in production on autonomous truck loading and unloading for national carriers, and own low-latency model deployment (distillation, QAT/PTQ, sub-millisecond inference on accelerators). Earlier I contributed to early Isaac Gym at NVIDIA and completed a CMU PhD on simulation-driven sensor design: 11 publications (AAAI, CoRL, ICRA, IROS, IJRR, Nature Communications Engineering) and 3 patents pending. Interested in post-training, evaluating, and deploying robot foundation models in the real world.

INDUSTRIAL EXPERIENCE

Dexterity — Redwood City, California

Senior Robotics Engineer, Lead GPU Simulation and Reinforcement Learning — Jul 2026 – Ongoing

- Lead RL post-training of online bin-packing policies running in production across multiple customer sites, improving pack density and throughput over the heuristic baseline.
- Cut instance-segmentation inference latency on an NVIDIA L4 GPU via distillation, quantization-aware training (QAT) and post-training quantization (PTQ) with minimal accuracy loss vs. the teacher; deployed to production robots.
- Lead neural stereo model training and distillation for real-time depth on production cells, delivering faster inference while preserving depth accuracy.
- Diagnose policy failures across perception, policy, control and hardware on live customer deployments; built tracing, profiling and benchmarking tooling used across the autonomy stack to find and remove compute bottlenecks.

Dexterity — Redwood City, California

Robotics Engineer III — Jul 2024 – Jun 2026

- Lead author, *Dexterity Supercharges Foresight by 17x with NVIDIA* (Mar 2026): cut world-model perception latency 17x (1508 ms to 90 ms) on an NVIDIA L4 via TensorRT (2x detection), GPU geometric reconstruction (4.6x on 32x more points) and custom CUDA physics kernels (94x), enabling sub-100 ms dual-arm trailer loading with 100+ tracked objects.
- Built a GPU-first RL pipeline (GPU physics with force and tactile sensing, thousands of parallel environments) that trained grasping policies and transferred them to real robots.
- Developed CUDA algorithms for motion planning and collision checking on the Accelerated Computing and Motion Planning team, reducing robot cycle time.
- Accelerated vision-model inference 10x (250 ms to 25 ms) for real-time deployment using TensorRT, FP16 and custom kernels; developed end-to-end 3D package detection for perception.

NVIDIA, Project Isaac — Pittsburgh, Pennsylvania

AI / Robotics Engineer — Jul 2018 – Aug 2019

- Contributed to an early version of **Isaac Gym**: built GPU-parallel environments and sim-to-real RL / optimal-control pipelines on the Simulation Technology Robotics team.
- Developed optimal-control and RL algorithms for legged locomotion and manipulation; sim-to-real on a Kinova Jaco arm and Minitaur quadruped.

Intel, Data Center AI — Santa Clara, California

Rendering Research Intern, GPU Research Group — May 2023 – Aug 2023

- Researched reinforcement learning for path guiding through transparent and glossy surfaces in physically based ray tracing.

EDUCATION

Carnegie Mellon University, School of Computer Science — Pittsburgh, Pennsylvania

PhD in Robotics — Aug 2019 – Jul 2024

- **Thesis**: A Modularized Approach to Vision-based Tactile Sensor Design Using Physics-based Rendering
- Built a physics-based light-transport simulation pipeline for vision-based tactile sensors (Mitsuba 0.6/2/3, NVIDIA OptiX 7, CUDA backend) used for sim-to-real sensor design; selected as a Nature Communications Engineering Editor's Choice 2025 (top 20 papers).

- Led a CMU–Siemens–Boeing defect-inspection project across two sites (IROS 2023, patent pending); co-instructed a tactile-sensor design course at UIUC.

Carnegie Mellon University, School of Computer Science — Pittsburgh, Pennsylvania
M.S. in Robotics — Aug 2016 – Jul 2018
- GPA 4.04 / 4.0. **Thesis**: Deep Reinforcement Learning with Skill Library: Exploring with Temporal Abstractions and Coarse Approximate Dynamics Models [pdf]. Built a multi-node MPI reinforcement-learning environment on MuJoCo for active-vision manipulation (CoRL 2018).

Indian Institute of Technology Kanpur — Kanpur, India
B.Tech. in Electrical Engineering — Jun 2012 – May 2016

PUBLICATIONS & PATENTS

- Vision-based Tactile Sensor Design Using Physically Based Rendering. Nature Communications Engineering, 2025 (Editor's Choice)
  Arpit Agarwal, Achu Wilson, Timothy Man, Edward Adelson, Ioannis Gkioulekas, Wenzhen Yuan
- A Modularized Design Approach for GelSight Family of Vision-based Tactile Sensors. International Journal of Robotics Research, 2025
  Arpit Agarwal, Amin Mirzaee, Xiping Sun, Wenzhen Yuan
- PneuGelSight: Soft Robotic Vision-based Proprioception and Tactile Sensing. International Journal of Robotics Research, 2025
  Ruohan Zhang, Uksang Yoo, Yichen Li, **Arpit Agarwal**, Wenzhen Yuan
- In Vivo Skin 3-D Surface Reconstruction and Wrinkle Depth Estimation Using Handheld High Resolution Tactile Sensing. Advanced Healthcare Materials, 2025
  Akhil Padmanabha, **Arpit Agarwal**, Catherine Li, Austin Williams, Dinesh K. Patel, et al.
- Scalable, Simulation-Guided Compliant Tactile Finger Design. RoboSoft 2024
  Yuxiang Ma*, **Arpit Agarwal***, Sandra Q. Liu*, Wenzhen Yuan, Edward Adelson (* equal contribution)
- Robotic Defect Inspection with Visual and Tactile Perception for Large-scale Components. IROS 2023
  Arpit Agarwal, Abhiroop Ajith, Chengtao Wen, Veniamin Stryzheus, Brian Miller, Matthew Chen, Micah K. Johnson, Jose Luis Susa Rincon, Justinian Rosca, Wenzhen Yuan
- Simulation of Vision-based Tactile Sensors Using Physics Based Rendering. ICRA 2021
  Arpit Agarwal, Timothy Man, Wenzhen Yuan
- Grasp Stability Prediction with Sim-to-Real Transfer from Tactile Sensing. IROS 2022
  Zilin Si, Zirui Zhu, **Arpit Agarwal**, Stuart Anderson, Wenzhen Yuan
- Improving Grasp Stability with Rotation Measurement from Tactile Sensing. IROS 2021
  Raj Kolamuri, Zilin Si, Yufan Zhang, **Arpit Agarwal**, Wenzhen Yuan
- Model Learning for Look-ahead Exploration in Continuous Control. AAAI 2019 (Oral)
  Arpit Agarwal, Katharina Muelling, Katerina Fragkiadaki
- Reinforcement Learning of Active Vision for Manipulating Objects under Occlusions. CoRL 2018
  Ricson Cheng, **Arpit Agarwal**, Katerina Fragkiadaki

Patents pending
- Multimodal Inspection of Large-scale Surfaces Using Vehicle-borne Sensors. US 18/753,819
- Continuous Surface Inspection Using a Roller Sensor. PCT/US2025/019008
- Methods, Systems, and Devices for Sensing (tactile 3-D skin reconstruction). US 63/865,982

Reviewer: ICRA 2018, 2020–23, 2027; CoRL 2026; IROS 2018–19; RA-L 2022–; T-RO 2023–

SKILLS

ML / RL: PyTorch, libtorch, TorchRL, RL (PPO, SAC, DDPG), imitation learning, distillation, QAT / PTQ
Simulation: Isaac Gym, Isaac Lab, MuJoCo, GPU physics, domain randomization, Mitsuba, OptiX
Systems: CUDA kernels, TensorRT, C++, Python, ROS, MPI multi-node training, profiling / benchmarking